Two Samples of Service Times

Unit 4 · Topic 4.7 · THE PROBLEM

Suppose two depots independently take SRSs of 36 weekday service-time records from monthly populations of 900 North and 1,200 South records. Technology gives $df = 69.31$ and $t^* = 1.995$ for a 95% two-sample interval.

Tariq: "I sorted both samples and paired equal ranks. The resulting differences have $\bar{d} = 2.2$ and $s_d = 3.0$, giving $(1.18, 3.22)$. Pairing makes the estimate more precise."

Mei: "The records were sampled separately, so I would use a two-sample interval. Sorting observed responses may change the uncertainty rather than reveal design-based pairs."

Service-time samples (min)

Depot	N	n	$\bar{x}$	s
North	900	36	42.3	8.4
South	1,200	36	40.1	7.6

FIRST TAKE — one sentence before you work the parts

Does sorting the times make the North and South records real pairs? Explain in one sentence.

DOK 1 **(a)** Find the North-minus-South difference in sample mean service times.

DOK 2 **(b)** The independent-sample standard error is 1.88797 minutes. Use $t^* = 1.995$ to calculate the 95% interval.

DOK 3 **(c)** In 3–4 sentences, choose a procedure using the original sampling design. Compare the two intervals and explain how sorting the times could mislead a reader about whether the population means differ.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.